

AGC iReady Data Dashboard

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About Nonprofit

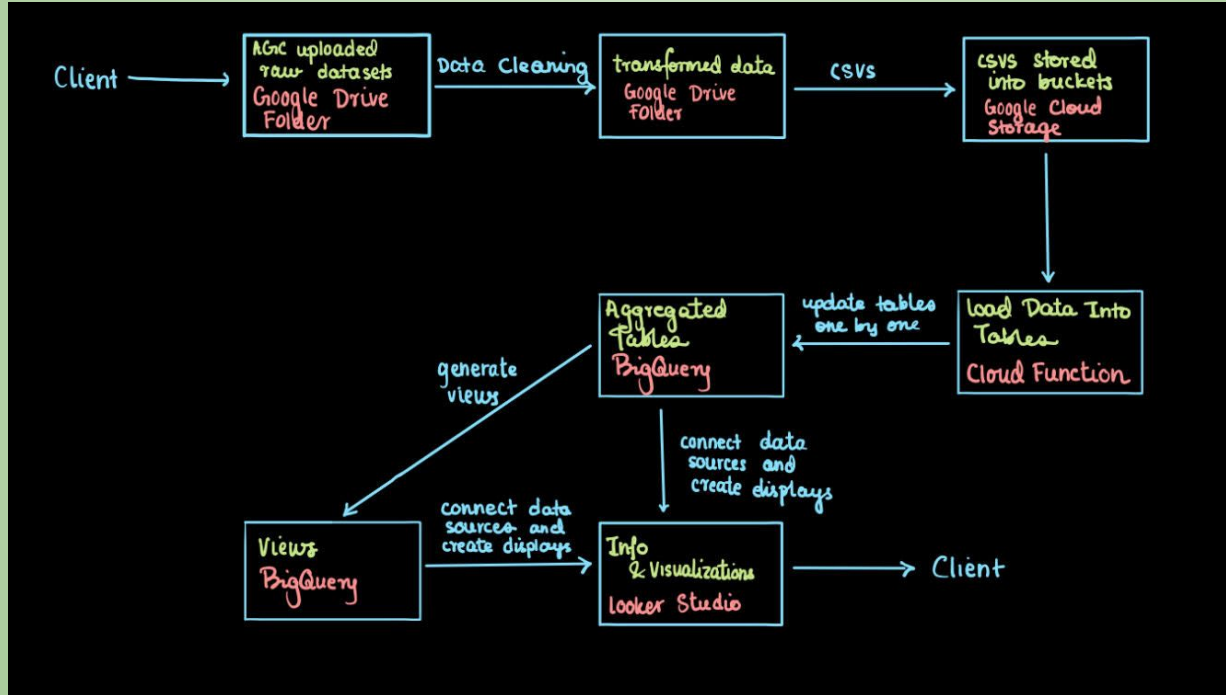
- Academy for Global Citizenship (AGC), a non-profit Chicago public charter school and incubator for educational innovation founded in 2005.
- Teacher training institute that has trained 10,000+ educators, impacting 5 million students globally.
- An accredited IB World School that emphasizes social justice, sustainability, entrepreneurship, dual language learning, and wellness.
- Serves students from early childhood to 8th Grade.

Project Goals

Building a dashboard with visualized student education data.

1. Let client to upload files into Math or ELA subfolder inside of a Google Drive
2. Import i-Ready data into Google Cloud Platform from Google Drive
3. Create a Database Schema to organize the i-Ready data
4. Create views organized i-Ready data for specific aggregation and calculations
5. Use views to create finalized visualizations

Process



Tech Stack

Infrastructure: Google Cloud Platform → Google Cloud Storage, Google BigQuery, Google Cloud Run, Google Looker Studio



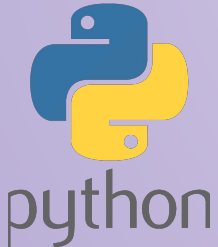
Google Cloud



Google
Big Query



Programming Languages: Python, SQL



Data Input and Cleaning

Sources

- Excel files (Math/ELA) are uploaded from Google Drive into an intermediate bucket in Google Cloud Storage (GCS).
- There are two separate GCS buckets—one for Math data and one for ELA data.

Triggers

- A Cloud Function ([transfer_files](#)) is invoked whenever a new file is detected in the intermediate bucket.
- That function:
 1. Downloads the Excel file from GCS.
 2. Reads in schema files (ELA/Math) from the schema folder in GCS.
 3. Cleans and converts the Excel data to CSV.
 4. Uploads the cleaned CSV to the appropriate final GCS bucket ([agc-iready-math-datasets](#) or [agc-iready-ela-datasets](#)).

Loading into BigQuery

- Another Cloud Function ([gcs_to_bq_agc_math](#) or [gcs_to_bq_agc_ela](#)) is triggered when the cleaned CSV appears in the Math or ELA bucket.
- It appends (or creates if not present) to the table in BigQuery ([combined_math_data_all_years](#) or [combined_ela_data_all_years](#)).

Data Cleaning and Preprocessing

Load Schema: Pull schema definitions from GCS (e.g. `ela_schema.csv` or `math_schema.csv`), which specify the correct column names and types.

Align Columns to Schema:

- Drop unneeded columns.
- Rename columns so they match BigQuery's field names.

Data Type Conversions:

- Convert floats to integers (replacing infinities, preserving nulls).
- Convert certain flags (`Y/N`) to boolean (`TRUE/FALSE`).

Output as CSV:

- The cleaned CSV is written into the final GCS bucket and then automatically loaded into the correct BigQuery table by the subsequent Cloud Function.

Views

- **Purpose:** Views are used to structure and simplify access to specific datasets within the converted tables, enabling targeted and efficient analysis for data visualizations
- **Examples:**
 - **student_ela_profile:** Contains individual student performance data in ELA for specific domains across year.
 - **student_math_profile:** Contains individual student performance data in Math for specific domains across years
 - **ela_domain_analysis:** Focuses on analysis of student performance within each ELA domain
 - **math_domain_analysis:** Focuses on analysis of student performance within each Math domain
- **Benefits:**
 - Simplifies querying for specific analysis without modifying raw data
 - Provides easy access to aggregated data, enabling faster visualizations
 - Supports the creation of specific reports and insights based on subject, domain, or cohort

Visualizations

- **Purpose:** Data visualizations provide actionable insights from the student assessment data by showcasing key trends and performance metrics
- **Examples of Visualizations:**
 - **ELA Performance Trends:** Visualized progress in ELA over time, showing trends in average scores per domain and cohort
 - **Math Performance Trends:** Visualized progress in Math over time, with a focus on individual student performance and overall cohort results
 - **Domain-Specific Analysis:** Graphs that break down student performance within individual domains for both ELA and Math
 - **Cohort Progression:** Progression charts that track student growth over the years, showing how cohorts are performing
- **Benefits of Visualizations:**
 - Enables stakeholders (teachers, administrators) to quickly interpret performance data
 - Highlights areas of strength and improvement
 - Facilitates decision-making for targeted interventions or curriculum adjustments

Dashboard Demo

Future

- **Scalability**
 - Handle more data with good performance
- **Machine Learning**
 - Predicting performance trends for specific student or group
 - Anomaly detection in student performance
- **Improved visualizations**
 - Increase interactivity (more filters, dropdowns, etc) and/or make the dashboard simpler to interact with
 - More visualizations depicting broader areas of iReady dataset
- **Security**
 - Encryption

Reflection

This project was an incredible learning experience!

- **Tech**

- Google Cloud Platform (Cloud)
- SQL syntax
- Data Pipeline components
 - i. Data Cleaning and Preprocessing
 - ii. Database (BigQuery)
 - iii. Data Visualization (Looker Studio)

- **Professional**

- Communicating with a client and reciprocating their needs
- Working in a team and subgroups
- Communicating progress, asking questions, and offering suggestions
- Project management (EM and PM)

Thank you!